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1. Use Case Description — Manage Inventory

Field	Description
Use Case Name	**Manage Inventory**
Actors	**Warehouse Staff (Primary)**; Manager (Secondary)
Preconditions	• User is authenticated and has the required permissions. • Inventory (Product/SKU and Stock Record) data exists in the system.
Normal Flow	**1.** The actor navigates to the Inventory module. **2.** The system displays a list of all products/SKUs with current stock across locations. **3.** The actor selects the required operation: Add, Update, Adjust, Delete, or Search. **4.** The system displays the relevant form or details. **5.** The actor enters or updates the required information. **6.** The system validates the entered information. **7.** The system saves the changes to the inventory database. **8.** If stock falls below the reorder point, the system triggers a low-stock alert. **9.** The system displays a success message and updates inventory details.
Postconditions	• Inventory data is created/updated/deleted successfully. • Stock levels and last-updated timestamp are updated.
Alternative Flows and Exceptions	**3a. Invalid Operation Selected:** • The system displays an error message and prompts the actor to select a valid operation. **5a. Invalid Data Entered:** • The system displays validation errors and does not save the changes. **7a. Database Update Failure:** • The system rolls back the transaction and displays an error message.

Non-functional Requirements	• **Performance:** Inventory operations should complete within 3 seconds. • **Security:** Only authorized users can perform inventory operations. • **Reliability:** System must maintain data consistency during updates. • **Usability:** Interface should be simple and voice-enabled for supported operations.

2. Use Case Description — Manage Purchase Order

Field	Description
Use Case Name	**Manage Purchase Order**
Actors	**Manager (Primary)**; Supplier (External)
Preconditions	• User is authenticated and has permission to manage purchase orders. • Products/SKUs and Suppliers exist in the system.
Normal Flow	**1.** The manager navigates to the Purchase Order module. **2.** The system displays the list of Purchase Orders. **3.** The manager selects **Create Purchase Order**. **4.** The system displays the PO creation form. **5.** The manager selects a supplier and adds products with quantities. **6.** The system calculates the total amount. **7.** The manager submits the PO for approval. **8.** The approver reviews and approves the PO. **9.** The system sends the approved PO to the supplier. **10.** When goods are received, the system updates stock and marks the PO as Received.
Postconditions	• Purchase order is created/approved and sent to supplier. • Upon receipt, stock levels are updated and PO status is updated.

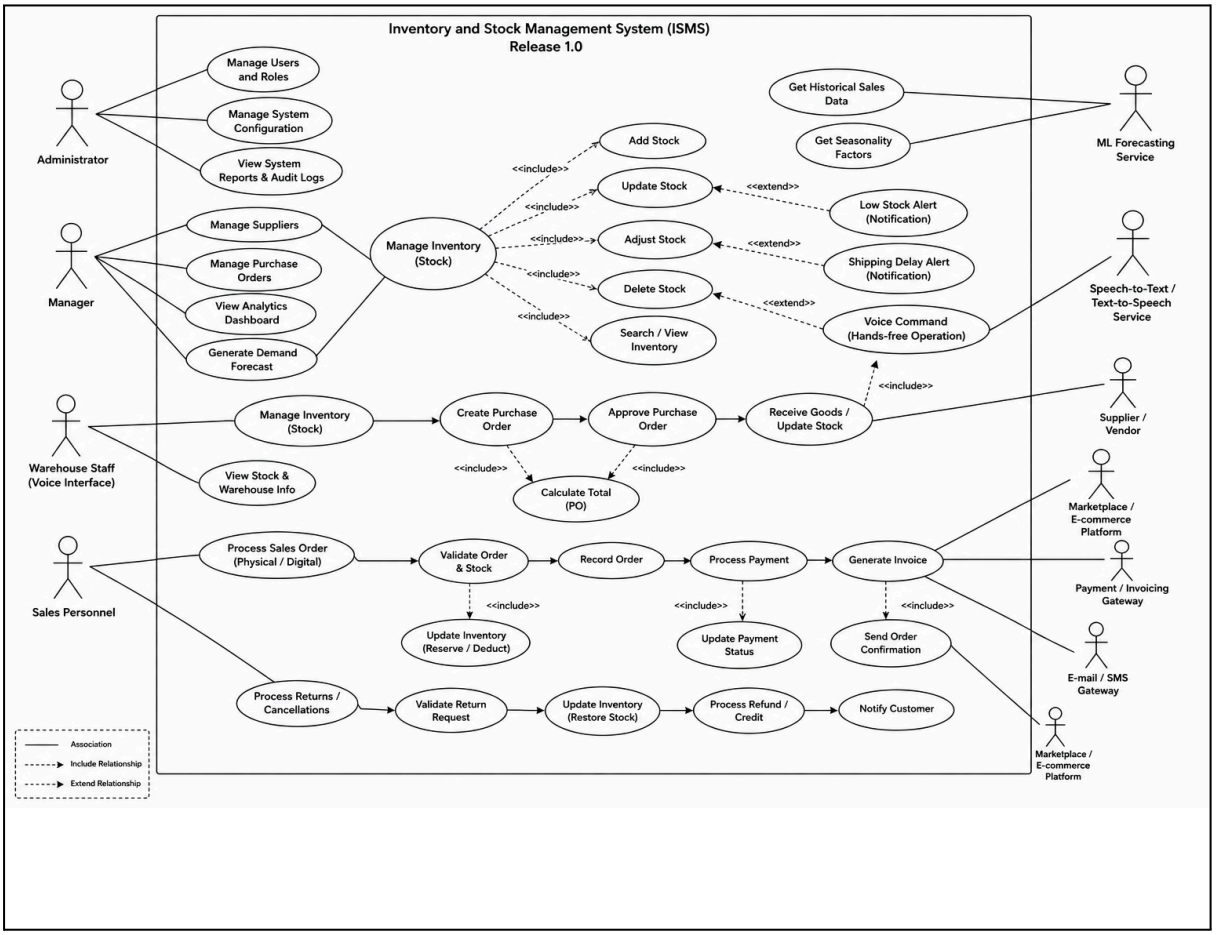
Alternative Flows and Exceptions	**5a. Product Out of Stock (in current inventory):** • The system shows current stock and recommended reorder quantity. **7a. Approval Rejected:** • The PO status is set to **Cancelled** and the manager is notified. **10a. Partial Receipt:** • The PO status is set to **Partially Received** and remaining items are pending. **10b. Receipt Failure:** • The system displays an error and does not update stock.
Non-functional Requirements	• **Performance:** PO operations should complete within 5 seconds. • **Security:** Only authorized users can create and approve POs. • **Reliability:** System must ensure accurate stock update on receipt. • **Auditability:** All PO actions must be logged.

3. Use Case Description — Process Sales Order

Field	Description
Use Case Name	**Process Sales Order**
Actors	**Sales Personnel (Primary)**; Customer (External)
Preconditions	• User is authenticated and has permission to process sales orders. • Products/SKUs are available in inventory.

Normal Flow	**1.** The sales personnel navigates to the Sales Order module. **2.** The system displays options to create a new order. **3.** The sales personnel enters customer details and adds products to the order. **4.** The system validates product availability. **5.** The system calculates the total amount. **6.** The sales personnel confirms the order. **7.** The customer makes payment (online or offline). **8.** The system processes the payment. **9.** The system generates invoice and order confirmation. **10.** The system deducts stock and updates inventory.
Postconditions	• Sales order is created and confirmed. • Payment is recorded, invoice is generated, and stock is updated.
Alternative Flows and Exceptions	**4a. Insufficient Stock:** • The system notifies insufficient stock and suggests alternate quantity or backorder. **7a. Payment Failure:** • The system does not confirm the order and allows retry of payment. **8a. Payment Gateway Unavailable:** • The system shows an error and allows selection of another payment method. **9a. Invoice Generation Failure:** • The system logs the error and allows retry.
Non-functional Requirements	• **Performance:** Order processing should complete within 5 seconds. • **Security:** Payment information must be encrypted and PCI-DSS compliant. • **Reliability:** System must ensure accurate payment and inventory update. • **Usability:** Support both online and offline sales scenarios.

Diagram:



Conclusion

The use case diagram for the Inventory and Stock Management System successfully represents the major interactions between the system, its users, and external entities. It identifies the key actors and their respective functionalities, while the <<include>> and <<extend>> relationships show how common and conditional activities are connected. The three detailed use case descriptions for Manage Inventory, Manage Purchase Order, and Process Sales Order further explain the normal flow, alternative flows, postconditions, and non-functional requirements of the system. Overall, the experiment helped in understanding and modelling the functional requirements of the Inventory and Stock Management System from the users perspective.